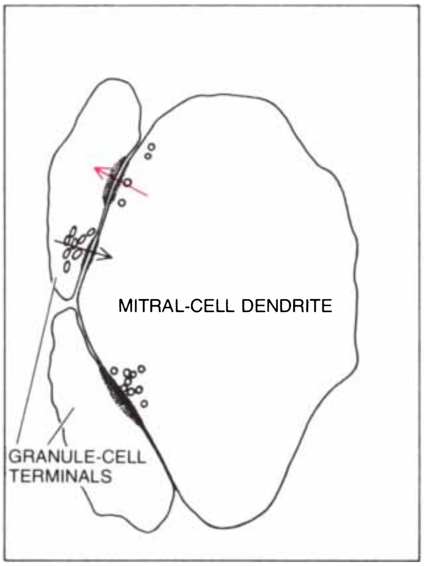
Microcircuits in the Nervous System

Nerve circuits are usually analyzed in terms of the axon, the long fiber of the nerve cell. It now appears that there are many circuits involving only the nerve cell's shorter extensions, the dendrites

by Gordon M. Shepherd

For nearly a century the investigation of the physiological basis of behavior has been dominated by the concept that the nervous system is composed of centers and pathways. Within the centers groups of neurons, or nerve cells, subserve specific functions such as the processing of sensory information and the control of movement. Some of these cells give rise to single long, thin fibers called axons, which transmit information from one center to another in the form of electrochemical impulses. Behavior therefore involves the continual transmission of impulses in the axons of different pathways.

This traditional view has naturally focused attention on the nerve impulse and the axon. Neurophysiologists have elucidated how the nerve impulse is generated and neuroanatomists have traced the trajectories of the axons. Such investigations have been brilliant chapters in the development of neuroscience. Yet they have left unanswered a central question: What controls the activity in the axons? The answer appears to lie in the intimate organization of the centers themselves. Recent studies of the connections and interactions between nerve cells within the centers—the microcircuits of the brain—have begun to provide the basis for a deeper understanding of the neural mechanisms underlying behavior.



SIMPLEST MICROCIRCUIT in the brain is the reciprocal synapse, a specialized junction between two nerve cells that transmits both excitatory and inhibitory signals. The electron micrograph on the opposite page, made by Joseph L. Price in the laboratory of T. P. S. Powell of the University of Oxford, shows reciprocal synapses between two types of nerve cells (mitral cells and granule cells) in the olfactory bulb of the rat. The large oval structure at the center of the micrograph is an extension of the mitral cell, seen in section; at the left are two terminals of the granule cell. The reciprocal synapse (*top left*) consists of two synaptic contacts that are side by side. The paired synapses have opposite polarities, that is, they conduct information in opposite directions. Moreover, one is excitatory (*colored arrow on map*) and the other is inhibitory (*black arrow*). The junction functions as an inhibitory feedback loop. The magnification is 77,500 diameters.

Dendrites and Synapses

Within each center, nerve cells give rise to dendrites: short, branching strands that terminate in the vicinity of the cell. The lengths and branching patterns of the dendrites are many and varied. Some dendrites branch sparingly, others profusely; some radiate in all directions, others are confined to single trunks or tufts. Dendrites may extend for less than a tenth of a millimeter or as far as several millimeters. The various types of dendritic branching pattern are characteristic of the nerve cells in each of the many specialized regions within the central nervous system.

In the 1950's the electron microscope revealed the intimate junctions that connect nerve cells: the synapses. At these junctions information is transferred from one neuron to another, usually by means of the chemical messengers known as neurotransmitters. A synapse between an axon and the soma (the cell body) of a neuron is called axosomatic; one between an axon and a dendrite is called axodendritic. For many years it was assumed that these were the only possible types of synapse and hence that the neuron was functionally polarized, receiving signals with its dendrites and cell body and transmitting them through its axon. As we shall see, the nervous system is much more resourceful in putting together synaptic circuits.

The effect of a synaptic input is to induce a postsynaptic potential: a slow change in the voltage across the outer membrane of the receiving neuron. Depending on the type of neurotransmitter involved, the postsynaptic potential is either excitatory or inhibitory. Whereas a nerve impulse traveling down an axon is an all-or-nothing phenomenon, the excitatory and inhibitory postsynaptic potentials are graded: the amplitude of the response depends on the intensity of the input. In neurons that have a long axon, if the sum of the excitatory potentials reaches a certain threshold, the cell will "fire," or send an impulse down its axon. If, on the other hand, the excitatory input does not reach the threshold, or if it is opposed by an inhibitory input, the cell will not fire. Thus the relative activities of the excitatory and inhibitory synapses impinging on a neuron determine the frequency of impulses transmitted through the cell's axon to other regions.

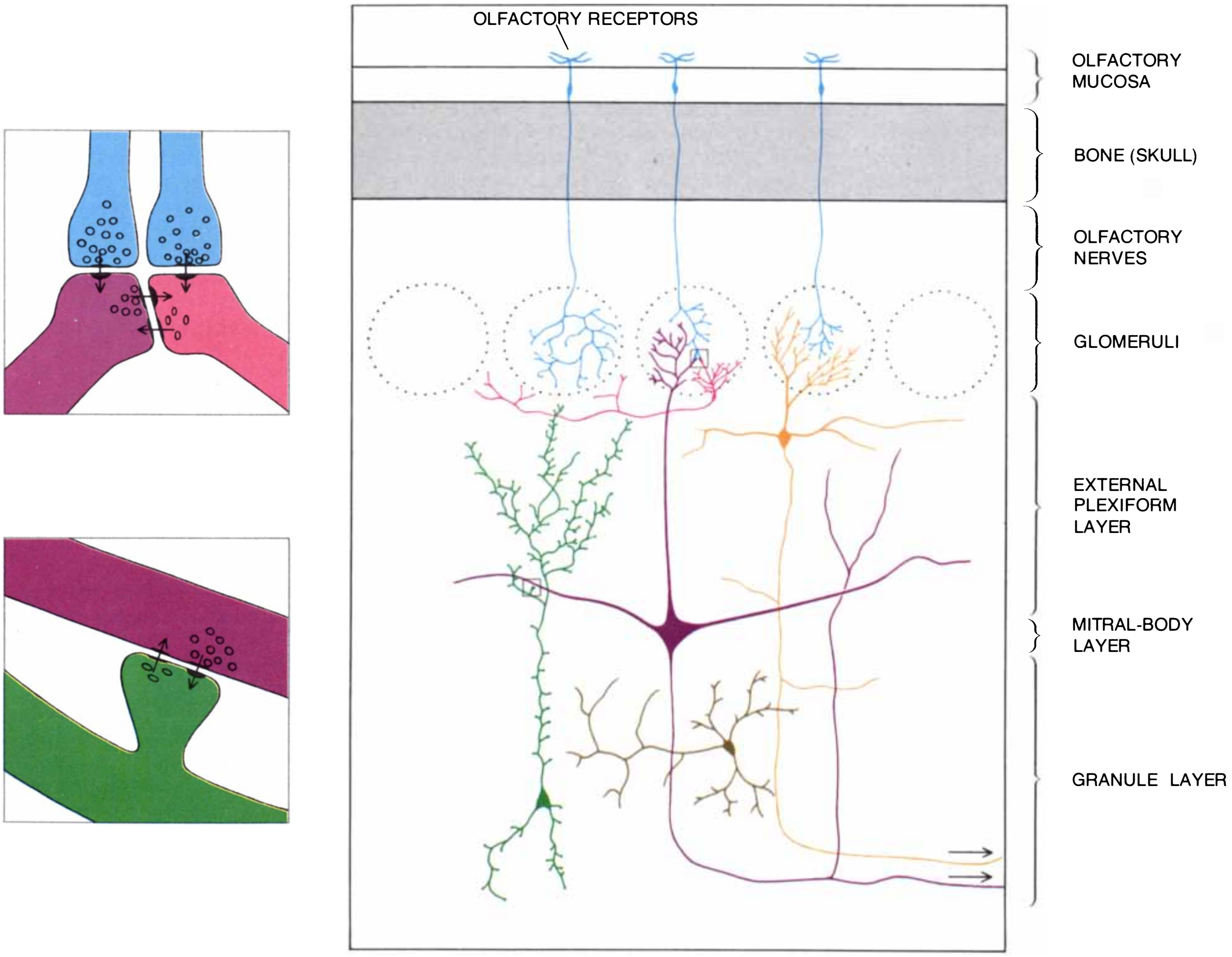
Investigation of synaptic inhibition showed that it can be mediated by axons projecting from neurons in distant centers or alternatively from the axons of neurons whose cell bodies are in the immediate vicinity. For example, the axon of a motor neuron in the spinal cord gives off a collateral branch that makes excitatory synapses onto a small short-axon cell called a Renshaw interneuron; the interneuron then makes inhibitory synapses back onto the motor neuron.

This inhibitory feedback loop, postulated by Birdsey Renshaw of Rockefeller University, was found in 1954 by John C. Eccles, Paul Fatt and K. Koketsu of the Australian National University. Since the Renshaw interneuron is confined to the same region of the spinal cord as the motor neuron it inhibits, the circuit is local compared with the long axonal pathways that connect one nervous center to another.

The Olfactory Bulb

In 1959, when I began my own research in the laboratory of Charles Phillips at the University of Oxford, the Renshaw inhibition pathway was the only type of local feedback circuit that had been well studied. Phillips was interested in determining whether similar circuits might be found in other parts of the central nervous system. To this end we, together with T. P. S. Powell, decided to study the organization of the neurons in the olfactory bulb, an oval structure attached to the front part of the brain that serves as a relay station between the olfactory receptors in the nose and the olfactory cortex in the brain.

The advantages of the olfactory bulb for neurobiological investigation are several. First, the input and output pathways of the bulb are separate: the axons from the olfactory receptor cells all enter the surface of the bulb, and the output axons arise from a different layer and leave through the depths of the bulb to connect to the olfactory cortex. This clear separation is quite rare in the nervous system; it is of great value to the experimenter, who can activate the pathways specifically. Second, the three major types of nerve cells in the olfactory bulb are arranged in discrete layers, which makes it possible to easily correlate anatomical and physiological studies. The principal neuron in the bulb is the mitral cell, which has a long axon projecting to the olfactory cortex. Its dendrites are of two types: branched secondary dendrites that extend laterally from the cell body and a long trunklike primary dendrite that terminates in a tuft of branches. The dendritic tuft of each mitral cell receives incoming axon terminals from olfactory receptor cells

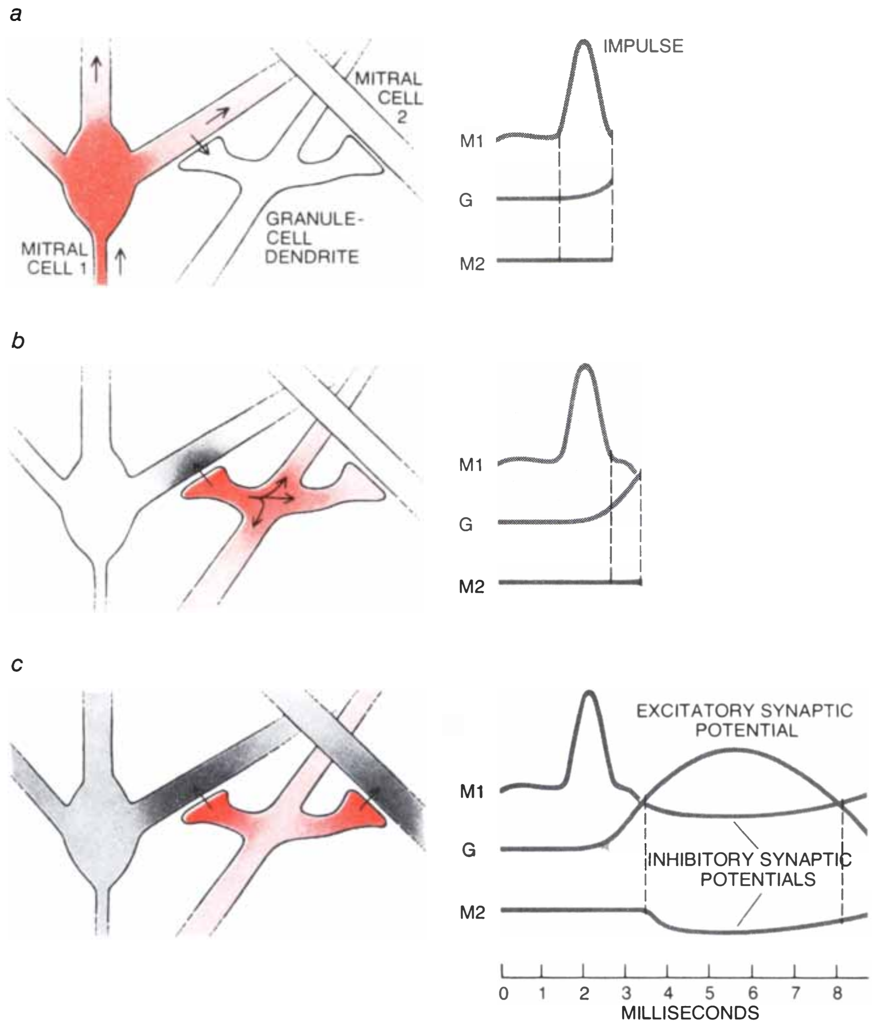


OLFACTORY BULB, a relay station linking the sensory receptors in the nose to the brain, contains numerous microcircuits that transmit and process olfactory information. In many of these circuits the nerve cells communicate directly through their dendrites (the short, branching extensions of the cell body) rather than sending an impulse down the axon (the nerve fiber). This diagram shows the various types of nerve cells in the olfactory bulb as revealed by treating the tissue by the Golgi method, which darkens only a few cells but does so throughout their extensions. The bulb is divided into discrete layers. Within the semispherical glomeruli the axons from the olfactory receptor cells (*blue*) make synapses with the dendrites of the periglomerular cells (*red*) and the dendritic tufts of the mitral cells (*purple*). Other types of neurons, including tufted cells (*orange*), also extend their dendrites into the glomerulus. In the external plexiform layer the horizontal secondary dendrites of the mitral cells make synapses with the "spines," or protruding terminals, that cover the dendrites of the granule cells (*green*). The deep granule layer of the olfactory bulb contains the output axons of the mitral cells, which project to the brain, and short-axon interneurons (*brown*). Insets at the left show main types of synaptic connections in glomeruli (*top*) and the external plexiform layer (*bottom*) as revealed by the electron microscope. Reciprocal synapses between dendrites are present in both regions.

within a small spherical region called a glomerulus, which acts as a kind of miniature relay station within the bulb.

The other two types of neurons in the olfactory bulb have all their processes confined within the bulb, and by that definition they are interneurons. One type, the periglomerular cell, sends a small dendritic tuft into a glomerulus and gives rise to a short axon that terminates nearby. The other type, the granule cell, sends out in opposite directions two long vertical dendrites that are covered with tiny cytoplasmic "spines." As long ago as 1875 the Italian neuroanatomist Camillo Golgi noted that the granule cell lacks an axon. This observation left the function of the cell entirely enigmatic: if, as the classical doctrine required, the output of a neuron is through its axon, what could be the output of a neuron without an axon?

Our initial studies showed that following an electrical stimulus to the axon of the mitral cell a nerve impulse is conducted along the axon to invade the cell body. (In this experimental situation the impulse travels in a direction that is the reverse of the normal one.) The spread of the impulse into the dendrites of the mitral cell is followed by a period of inhibition. Although the impulse lasts for only a millisecond or two, the subsequent inhibition lasts for up to several hundred milliseconds, which is quite long on the time scale of neurophysiological events. We hypothesized that the inhibition of the mitral cell was due to the action of the granule cell. How a cell without an axon could function as an inhibitory interneuron, however, remained a tantalizing mystery.



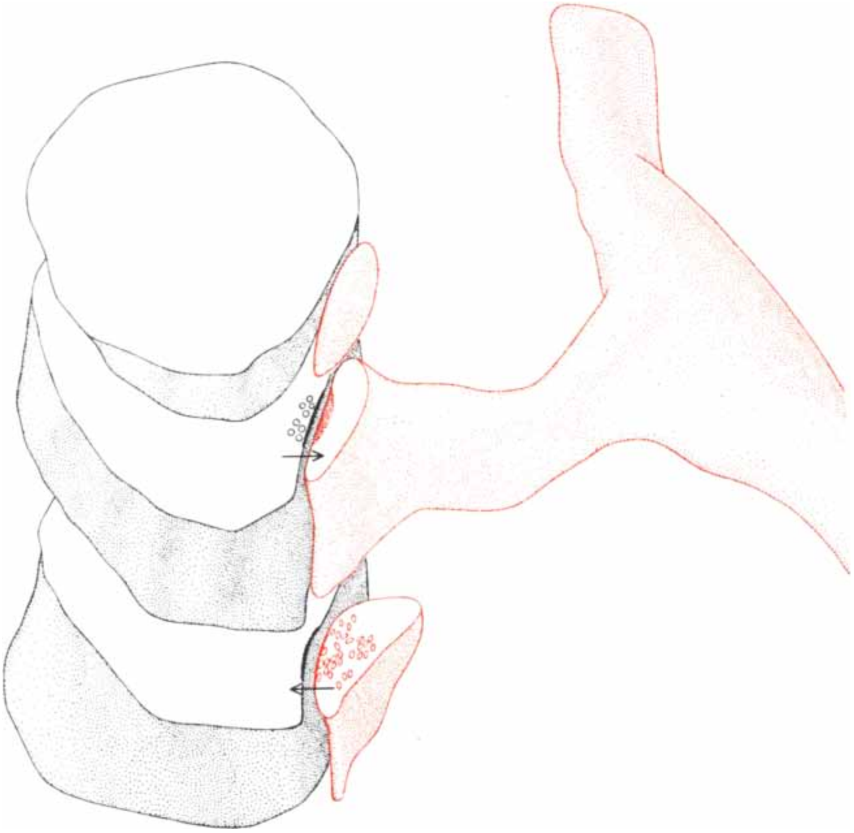
FUNCTION OF THE MICROCIRCUIT between the mitral cell and the axonless granule cell in the olfactory bulb was deduced by interpreting the results of physiological experiments with the aid of computer simulations. Here the activity in the circuit during a typical experiment is diagrammed. First the mitral-cell axon is electrically stimulated, triggering an impulse that is conducted backward up the fiber to the cell body. The impulse travels throughout the secondary dendrites of the mitral cell, activating excitatory synapses onto the dendritic spines of the granule cell (*a*). The excitation of the granule cell in turn activates inhibitory synapses from the granule-cell dendrite back onto the mitral-cell dendrite (*b*). As a result of this feedback loop the initially excited mitral cell is inhibited from firing. Moreover, because of the passive spread of current in the granule-cell dendrite, inhibitory synapses in other parts of the dendrite act to inhibit additional mitral cells (*c*). Normally, of course, the initial impulse in the mitral cell is triggered not by electrical stimulation but rather by input from the olfactory receptor cells.

Dendrodendritic Synapses

The mechanism of action of the granule cell was still unknown in 1962, when I moved to the laboratory of Wilfrid Rall in the Mathematical Research Branch of the National Institute of Arthritis, Metabolism, and Digestive Diseases. Rall was one of the first to recognize the importance of dendrites in the synaptic integration of neuronal activity, and his most important finding was that the electric currents set up by synapses onto the dendrites flow throughout the dendritic tree. These "electrotonic" currents are the functional link between the sites of synaptic input on the dendrites and the site of impulse generation, if there is one, in the cell body of a neuron. Rall showed that the electrotonic currents and the potential changes associated with them could be rigorously described by taking account of the electrical properties of the dendrites and the geometry of their branching. His methods have now become the accepted basis for analyzing and understanding synaptic integration in dendrites throughout the nervous system.

Our work together was mainly concerned with constructing computer models of the mitral and granule cells that realistically represented their electrical and geometric properties. Beginning with the impulse conducted backward up the mitral-cell axon and into the cell body, we modeled the spread of activity in the dendrites of the mitral and granule cells, modifying our simulation as we went along so that it agreed with the physiological recordings Phillips, Powell and I had obtained at Oxford. The sequence of events that fit the experimental data was an unexpected one. According to our computer model, the arrival of an impulse in the mitral cell triggers the activation of excitatory synapses from the secondary dendrite of the mitral cell onto the spines of the dendrite of the granule cell. This excitatory synaptic input depolarizes the granule cell (that is, it discharges the voltage across its outer membrane) and activates inhibitory synapses back onto the mitral cell to produce the observed long-lasting inhibition. The graded depolarization is conducted electrotonically throughout the dendritic tree of the granule cell, so that inhibitory synapses onto neighboring mitral cells are activated as well.

This scheme was highly unorthodox



RECONSTRUCTION OF A RECIPROCAL SYNAPSE in three dimensions shows a granule-cell spine (*color*) forming a junction with the secondary dendrite of a mitral cell. Within the single ending are two synaptic contacts with opposite polarities. The synapse from the mitral cell to the granule cell (*top*) has round vesicles and a fuzzy thickening of the postsynaptic membrane characteristic of an excitatory synapse, whereas the synapse from the granule cell to the mitral cell (*bottom*) has flattened vesicles characteristic of an inhibitory synapse. The reconstruction was done by Thomas S. Reese and Milton W. Brightman at the National Institute of Neurological Diseases and Stroke by tracing a series of electron micrographs of 23 consecutive sections through the junction; no sections are omitted in showing the cut surfaces.

in that it postulated the existence of inhibitory and excitatory synaptic interactions between dendrites, for which there was no precedent at the time. By good fortune, however, we were in frequent contact with Thomas S. Reese and Milton W. Brightman, who were studying the structure of the olfactory bulb with the electron microscope in a nearby laboratory at the National Institute of Neurological Diseases and Stroke. It was an exciting moment early in 1965 when we learned that synapses between the dendrites of mitral and granule cells actually exist. The discovery of such dendrodendritic synapses, together with our functional model, contradicted the classical doctrine that the nerve cell could only receive signals with its dendrites and cell body and transmit them through its axon, since it suggested that neurons can communicate with each other through their dendrites without the intervention of an axon or a nerve impulse.

The careful studies of Reese and Brightman made it possible to reconstruct the dendrodendritic synapses in the olfactory bulb in three dimensions and to clearly identify them with the dendrites of the mitral and granule cells. The most striking aspect of the structure was that the excitatory and inhibitory synapses were arranged in reciprocal pairs oriented in opposite directions, and hence they were appropriate for mediating the postulated sequence of an excitation proceeding from the mitral cell to the granule cell followed by an inhibition proceeding from the granule cell to the mitral cell.

Anthony Pinching and Powell at Oxford and Edward White in Reese's laboratory subsequently found that the short-axon periglomerular cells also make connections with the dendritic tuft of the mitral cell inside the glomerulus through reciprocal dendrodendritic synapses. Moreover, in my laboratory at the Yale University School of Medicine, Thomas V. Getchell and I have obtained physiological evidence that the periglomerular cells are inhibitory in their synaptic actions. Thus the two types of interneurons in the olfactory bulb share basic features in their interactions with the mitral cells.

For many research workers the existence of dendrodendritic circuits has been difficult to reconcile with traditional notions of neural organization based on circuits formed by axons. The dendrodendritic circuits have been termed "primitive," "nonusual" and "unconventional." The available evidence indicates, however, that dendrodendritic synapses are similar in basic respects to the synapses made by axons. It is now apparent that they are a logical and economical way to organize synaptic interactions in the minimum space. The reciprocal synapse between dendrites is the most compact synaptic circuit yet identified in the nervous system. We can therefore consider it to be a microcircuit, at the opposite extreme from the macrocircuits between nerve centers made through long axonal pathways.

Sensory Functions

How is the operation of microcircuits through dendrites related to the sensory function of the olfactory system? One important property of sensory systems is their sensitivity. We and others have found that molecules of odorous substances can be detected at extremely low concentrations in the air, a finding that implies a high degree of sensitivity in the circuits of the olfactory bulb that transmit under these conditions. Actually the situation is somewhat more complex. Computer simulations I have recently carried out in collaboration with Robert K. Brayton of the Thomas J. Watson Research Center of the International Business Machines Corporation suggest that different microcircuits of the olfactory bulb differ markedly in the sensitivity of their synapses to electrotonic current flow. For example, the synapses made by the secondary dendrites of the mitral cell appear to have a relatively low sensitivity, since they are activated by the spread of the impulse generated in the cell's primary dendrite by input from the olfactory receptor cells. In contrast, the synapses made by the granule-cell dendrites seem to have a relatively high sensitivity, since they are activated by graded synaptic potentials. High-sensitivity synapses may also be present in the olfactory glomeruli. More needs to be known about the electrotonic properties of dendrites, however, before the sensitivities of the dendrodendritic synapses in the olfactory bulb can be characterized quantitatively.

Recently John S. Kauer and I have been analyzing the responses of mitral cells in the olfactory bulb of the salamander to various olfactory stimuli, using pulses of odorous substances at carefully controlled levels of concentration. It is clear from our work so far that inhibition is important in shaping the responses of the mitral cells, and we are currently trying to identify the specific contributions of the circuits through periglomerular and granule cells to this inhibition. It seems safe to say that the ability to distinguish one odor from another depends on interactions between excitatory and inhibitory activity in the dendritic circuits of the olfactory bulb.

An entirely different approach has been to correlate the activity of the circuits in different parts of the olfactory bulb with different olfactory stimuli. To this end Frank R. Sharp, Kauer and I have utilized a new biochemical technique developed by Louis Sokoloff and his associates at the National Institute of Mental Health. Because nerve cells that are physiologically active consume glucose for fuel, they can be identified by providing them with a chemical derivative of glucose that is taken up like glucose by active cells but that cannot be further metabolized; the substance is therefore trapped in the tissue. The glucose derivative is labeled with atoms of a radioactive isotope, so that its location is revealed by its radioactivity. The radioactivity is detected by slicing the tissue into thin sections and placing them on photographic film; after a week's exposure the radioactive cells have darkened the emulsion of the film in proportion to their physiological activity.

With the aid of this technique we have found that odor stimulation is associated with spatial patterns of activity in the olfactory bulb. The foci of activity are located precisely in groups of glomeruli, which have a high density of dendrites and synapses. In collaboration with William B. Stewart and our colleagues in the Neurosurgical Research Laboratories at Yale we are currently analyzing the distribution of these active sites in relation to different odors, including pheromones, the chemical substances that coordinate social activity in insects and mammals. The glucose method is one of a growing battery of biochemical techniques available to neuroscientists, and it promises to be of great service in correlating synaptic organization with sites and levels of functional activity.

Distribution of Microcircuits

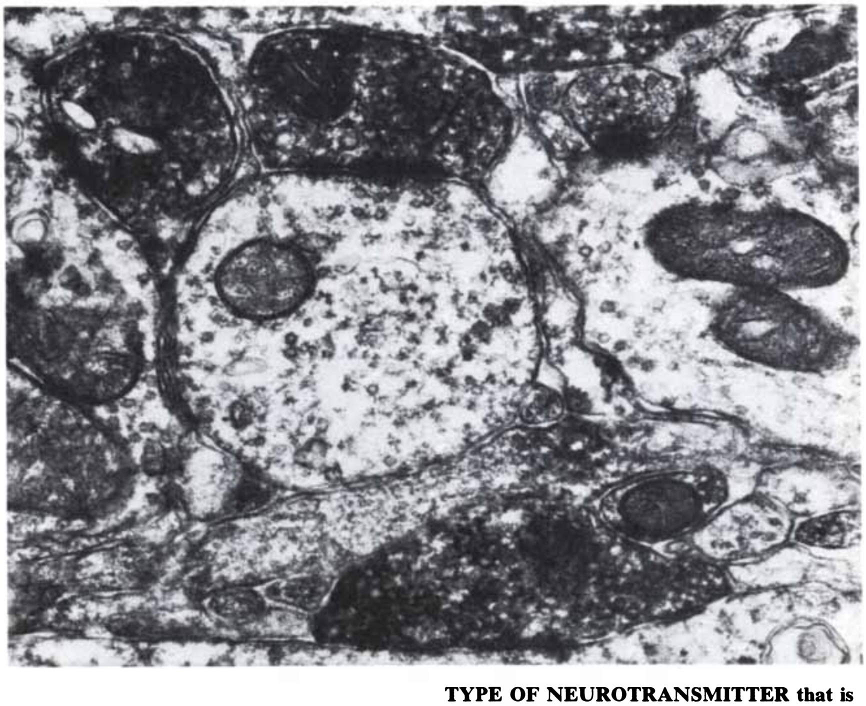
Are dendrodendritic synapses perhaps found only in the olfactory bulb? This possibility never arose, because as we were studying the bulb independent investigations by John E. Dowling, who was then at Johns Hopkins University, in collaboration with Brian B. Boycott of the University of London, showed the presence of similar synaptic arrangements in the retina of the eye. One type of interneuron in the retina, the amacrine cell, was known to lack an axon, and in this respect it presented the same puzzle that the granule cell in the olfactory bulb did. Dowling and Boycott helped to solve the puzzle by finding reciprocal synapses between the dendrites of the amacrine cells and those of neurons known as bipolar cells. Subsequent work by Frank S. Werblin and Dowling in 1969, and by many others since then, has shown that most of the cells in the retina generate slow synaptic potentials serving as both the synaptic response and the synaptic output, a situation analogous to the ones we have seen in the olfactory bulb.

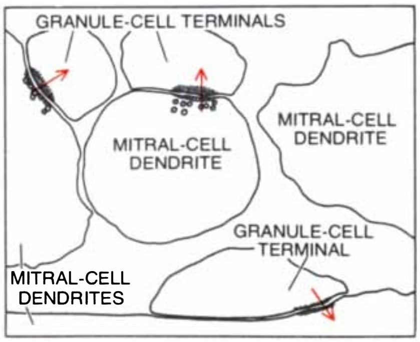
Over the years since the early work on the olfactory bulb and the retina much evidence has accumulated for similar types of synaptic circuits through dendrites in other parts of the nervous system. The evidence was slow to come at first; most neuroanatomists were trained to think of synapses as being made only by axons onto cell bodies or dendrites, and they had to retrain themselves (and in some cases revise earlier conclusions) to include synapses made by dendrites. Now, however, the list of synaptic circuits through dendrites is quite long, and it can be said that such circuits have been found in most of the major parts of the nervous system.

For example, the synaptic connections of the trigeminal nerve to the brain stem have been studied by Stephen Gobel at the National Institute of Dental Research. This nerve carries sensory information from the face, and it feeds into a complicated microcircuit that appears to be essential for integrating the information and transferring it to the brain. Several types of synaptic arrangements have been identified in this region: single and reciprocal axodendritic synapses, single dendrodendritic synapses and even axoaxonic synapses, in which an axon receives synapses from another axon. This variety of synaptic arrangements suggests the existence of many small computational units that provide for a highly localized and specific processing of information.

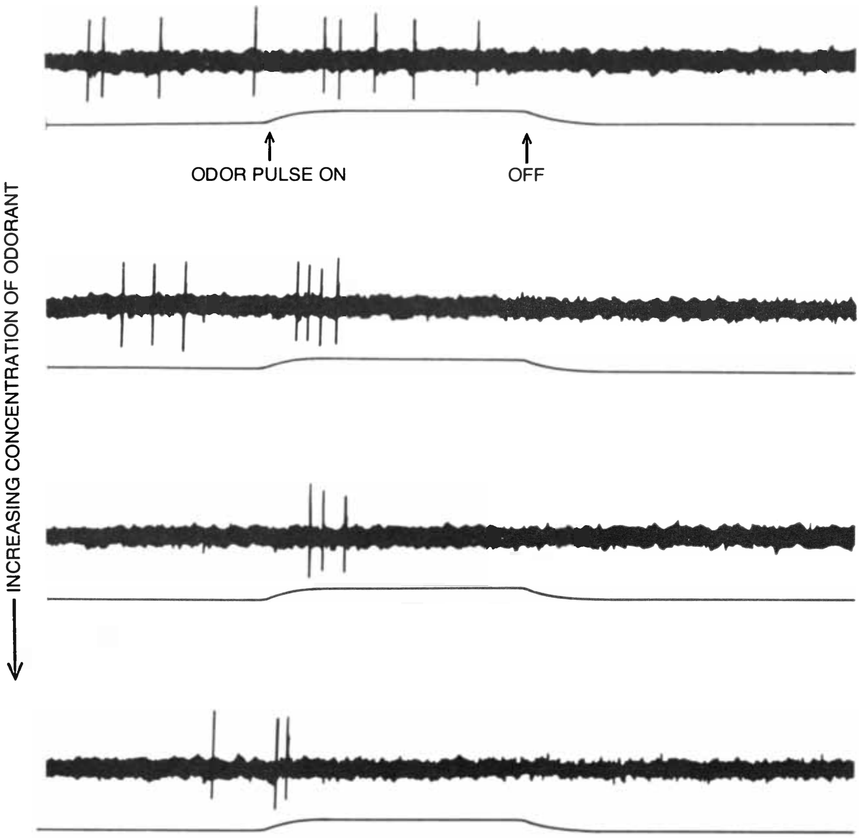
One of the important sensations mediated by the trigeminal nerve is that of pain. There is currently an enormous range of investigations of pain, including studies of the neuronal mechanisms of normal pain, of the dysfunctions responsible for states of intractable pain, and of the localization of opiate receptors in the nervous system. It is likely that our understanding of such problems will require precise knowledge of the dendritic circuits that process the inputs from the spinal and trigeminal nerves.

Another brain region in which dendritic circuits have been identified is the thalamus, the final relay center for sensory pathways to the cerebral cortex.





TYPE OF NEUROTRANSMITTER that is released at the inhibitory synapse from the granule-cell dendrite to the mitral-cell dendrite was identified by a cytochemical study conducted by Charles E. Ribak, James E. Vaughn, Kihachi Saito, Robert Barber and Eugene Roberts at the City of Hope National Medical Center in Duarte, Calif. In this electron micrograph the terminals of the granule cell contain a precipitate: a positive test for the enzyme glutamate decarboxylase, which converts the amino acid glutamic acid into the neurotransmitter gamma-aminobutyric acid (GABA). This finding suggests that GABA is released from the granule cell. Magnification of the micrograph is some 37,000 diameters.



PATTERNS OF IMPULSES generated by mitral cells in the olfactory bulb in response to pulses of an odorous substance were recorded with an extracellular microelectrode by John S. Kauer and the author at the Yale University School of Medicine. The substance they used was amyl acetate, which to human beings smells like bananas. (The odor pulse is represented by the lower trace, beginning at the arrow.) At a low concentration of amyl acetate (*top*) there was an impulse discharge in the mitral cells that lasted through most of the odor pulse. At higher concentrations the impulses fired faster and the discharge was cut short by a long-lasting inhibition. This interplay of excitation and inhibition appears to underlie processing in olfactory bulb.

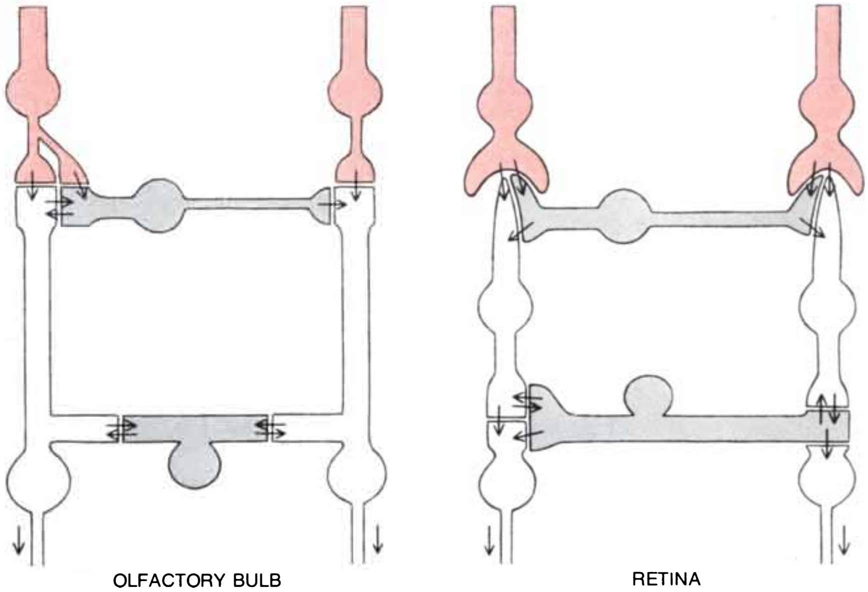
AUTORADIOGRAPH OF A THIN SECTION through the olfactory bulb reveals the focal activity of the synaptic circuits in response to a specific odor. To make this image Frank R. Sharp, Kauer and the author injected a rat with a radioactively labeled sugar, deoxyglucose, and then exposed the animal to air containing a vapor of amyl acetate. The nerve cells in the bulb that were most active in detecting and discriminating the odor preferentially took up the radioactively labeled sugar from the blood. When sections of the bulb were placed on X-ray film, the nerve cells darkened the film in proportion to their radioactivity, providing an activity map of the bulb. Here the darkest regions are located precisely over the olfactory glomeruli.

Within the thalamus incoming axon terminals make synapses onto the dendrites of both long-axon relay neurons and interneurons; the interneurons then make both dendrodendritic and axodendritic synapses onto the relay neurons. It therefore appears that much of the information flowing into the cerebral cortex is processed by microcircuits at the thalamic level. Dendrodendritic synapses have also been found in regions that regulate movement in monkeys, such as the motor area of the cerebral cortex and the basal ganglia of the cerebrum and the midbrain, which have been implicated in the derangements of movement that occur in Parkinson's disease. These regions are difficult to study, however, and the quantitative importance of dendrodendritic and other types of circuits found in them has yet to be established.

A final example is the suprachiasmatic nucleus, a tiny region in the forward part of the hypothalamus above the optic chiasm (the region where the optic nerve from one eye meets the optic nerve from the other). Fritz H. Güldner and J. R. Wolff of the Max Planck Institute for Biophysical Chemistry in Göttingen have shown that dendrodendritic synapses are a distinctive feature of the synaptic organization of the suprachiasmatic nucleus. Reciprocal synapses have also been observed there. What little is known about the region suggests that it plays an important role in slow cyclic behavior, such as the daily physiological cycles of the body, by controlling the release of certain hormones. It therefore appears that dendrodendritic synapses may be related not only to rapid information processing but also to slow activity measured in hours or days.

Invertebrate Systems

So far I have focused on the nervous system of vertebrate animals. In invertebrate animals information recently obtained suggests a striking parallel with the findings in vertebrates. The nervous system of invertebrates is characteristically organized in a chain of ganglia, or small nerve centers. Within a ganglion the nerve-cell bodies lie at the periphery; the dendrites arise from the axon and branch and intermingle in the center of the ganglion, at some distance from the cell bodies. This separation has greatly complicated the identification of the cells from which the dendrites arise. Recently, however, David King of the University of California at San Diego has been able to reconstruct the dendritic terminals within the ganglion of the lobster and has found that a given terminal can have both sending and receiving synapses. These results and others appear to confirm that the basic types of dendrodendritic synapses are present in



SIMILARITY BETWEEN THE SYNAPTIC ORGANIZATION of the olfactory bulb and the retina is shown in this schematic diagram. Both regions are way stations between the sensory receptor cells (*color*) and the brain. Both have a framework of vertical pathways (*open*) for straight-through transmission combined with horizontal pathways (*shaded*) for lateral interactions. The horizontal connections are organized in two main layers and involve single and reciprocal synapses that operate by combinations of slow graded potentials and impulse activity.

the invertebrate nervous system, and that some of the patterns of interconnection are similar to those found in the vertebrate nervous system.

Neurophysiologists have often taken advantage of the relative simplicity of the nervous system of invertebrates to explore the fundamental properties of nerve cells. As early as 1959 the electrophysiological activity of a number of different types of neurons in invertebrate ganglia had been recorded by impaling individual cells with thin glass electrodes. In reviewing these studies Theodore H. Bullock, who was then at the University of California at Los Angeles, concluded that the normal functions of many of the small neurons in the invertebrate ganglion might be mediated by graded potentials rather than by the generation of all-or-nothing nerve impulses. Evidence to support this hypothesis was difficult to obtain in invertebrate neurons because of the separation between the cell body, where the recordings are made, and the dendritic terminals, where the synapses occur. Recently, however, several examples of neurons have been found that communicate only through graded potentials.

Particularly striking are the results obtained by Keir Pearson and C. R. Fourtner at the University of Alberta with the nervous system of the cockroach. The legs of the cockroach arise from the thorax, and walking is controlled by alternating bursts of impulses in the nerves from the thoracic ganglia to the flexor and extensor muscles in the legs. The rhythmic and reciprocal nature of this activity arises in the ganglia themselves, and Pearson and his colleagues have shown that nonfiring neurons there are responsible for generating the motor pattern [see "The Control of Walking," by Keir Pearson; Scientific American, December, 1976].

Other Local Interactions

In this account of microcircuits I have emphasized dendrodendritic interactions, but I could also note a number of locations where interactions between axons (axoaxonic synapses) have been found or suspected. Similarly, I have focused on the role of graded potentials in microcircuits, but all-or-nothing nerve impulses do occur in short axons, in axon collateral branches and in some dendrites. In addition, although most synapses operate by releasing chemical neurotransmitters, there are a number of specialized regions called gap junctions where transmission is mediated by the direct flow of electric current. Finally, there are interactions between neurons that take place independently of specific points of contact; they include the continual flow of substances within nerve cells and across their membranes, and the electric fields that are set up when populations of nerve cells are active in concert. All these phenomena contribute to functional operations within the microenvironments of the nervous system.

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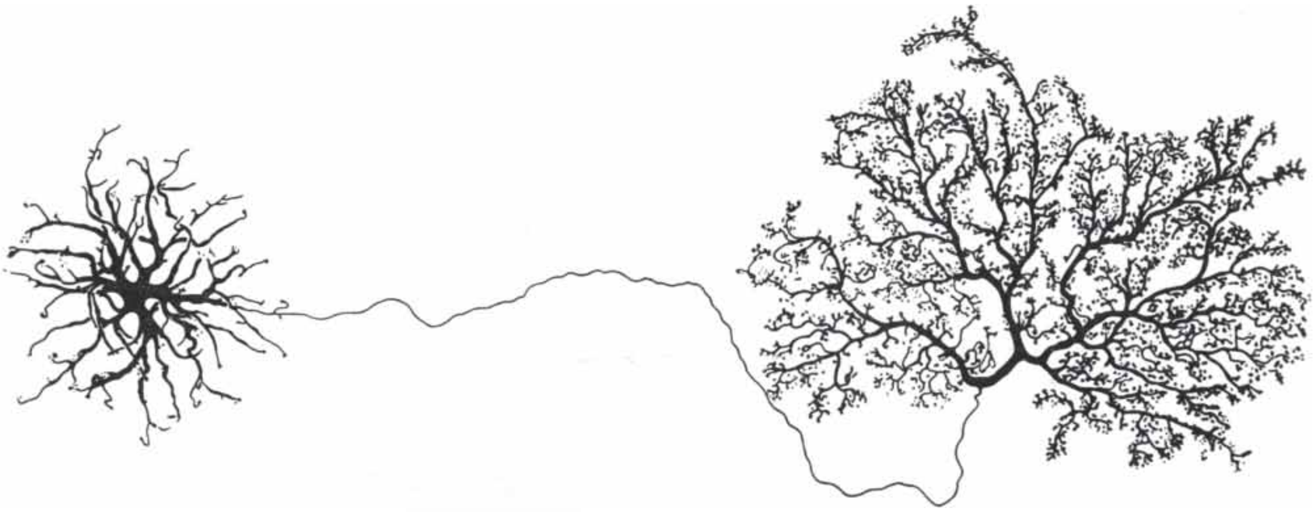
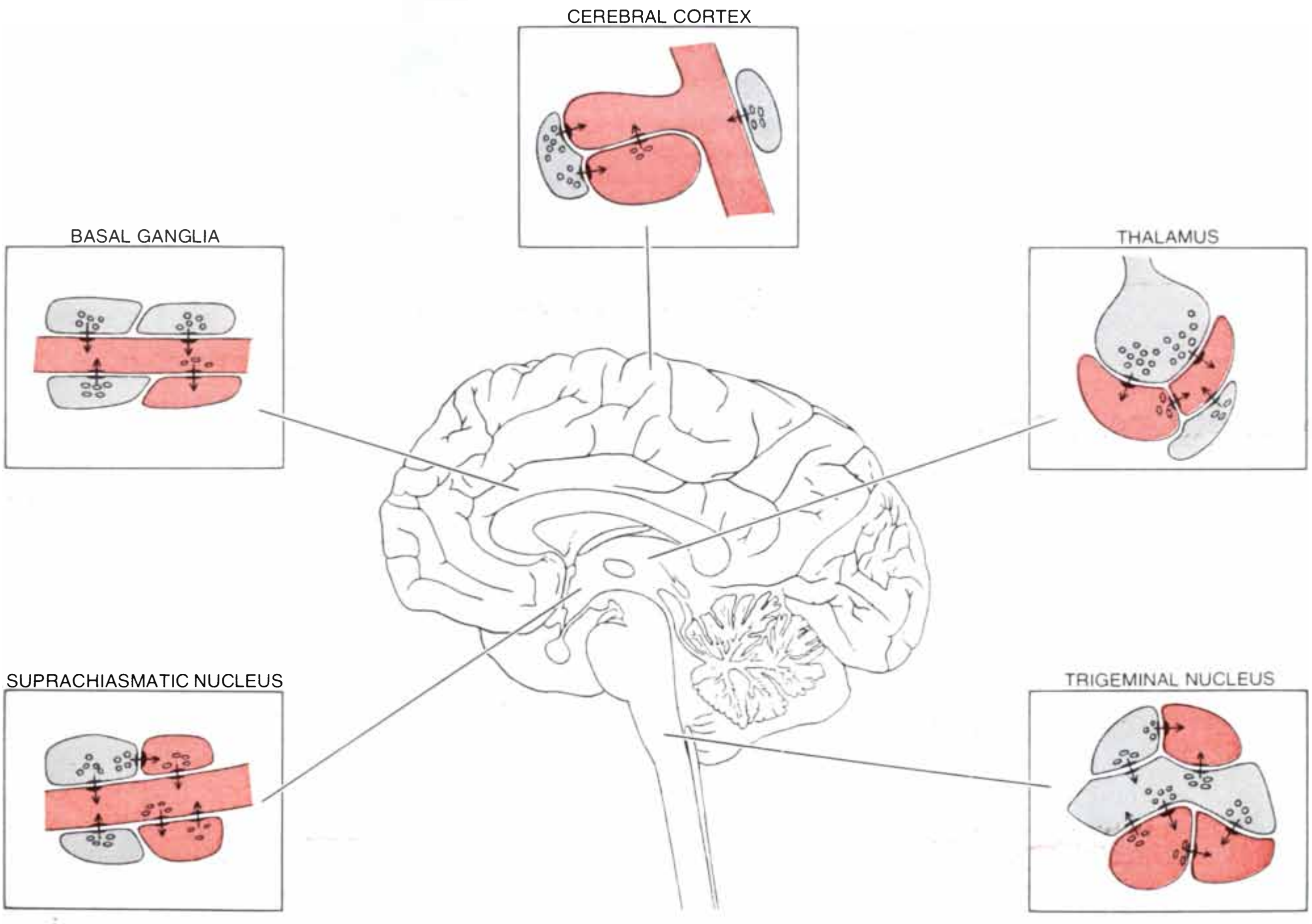


HORIZONTAL CELL FROM THE RETINA of a cat was visualized with the Golgi method by Brian B. Boycott of the University of London. The cell is divided into two highly branched regions connected by a thin fiber. Dendrites associated with the cell body (*left*) receive signals predominantly from the color receptor cells (cones), whereas the terminal arborization (*right*) receives signals primarily from the light-intensity receptor cells (rods). The axonlike process connecting these two regions neither generates impulses nor allows significant passive conduction between them. Instead it appears to serve to electrically isolate the cell body from its terminal arborization while maintaining a nutritive link. In some mammals this arrangement apparently restricts the interactions between the rod and cone systems, which have different dynamic ranges of function. The advantage of having single nerve cells with electrically isolated regions is that the number of independent integrating units within the brain can be increased without the need for augmenting number of individual cells.



TYPICAL MICROCIRCUITS in different regions of the brain are diagrammed. The insets represent in schematic fashion the outlines of terminals and the sites of synapses; dendrites are shaded in color and axons in gray. The synaptic connections in the thalamus and trigeminal nucleus are organized in clusters as shown; the connections in the other regions are more diffusely distributed than is indicated here. The circuits are involved in functions ranging from rapid processing in the thalamus to the generation of long-term rhythms.

Neurosciences Research Program, an interdisciplinary organization that conducts scientific meetings to explore current problems in the neurosciences, the new work on synaptic organization was reviewed. In this session, organized by Francis O. Schmitt of the Neurosciences Research Program and chaired by Pasko Rakic of the Harvard Medical School, the term "local circuits" was used to describe the pathways confined within nerve centers. The pathways through dendrites that occur in the olfactory bulb and other regions may be regarded as the prototypes of local circuits, characterized as they are anatomically by local connections and physiologically by local potentials.

Microcircuits and Behavior

How are investigations at this microscopic level of nervous organization related to the large-scale actions that characterize behavior? The simplest types of microcircuit are concerned with the properties of synaptic excitation and inhibition and with sequences of these activities. In many instances, as in the olfactory bulb or the retina, the patterns in which the synapses are organized are complicated but stereotyped, which suggests that some microcircuits might be organized into more complicated microprocessing units, each with a particular operation to contribute to local integration.

From considerations such as these it is becoming evident that the nervous system is built up of hierarchies of functional units of increasing scope and complexity. The traditional concept of the single neuron, receiving information by way of its dendrites and sending it out through its axon, can now be seen to represent only one type of functional unit within these hierarchies. The new findings indicate that a single neuron may comprise many functional units in terms of its individual synaptic relations. In addition any given neuron is only one small component of larger functional units made up of multineuronal assemblies.

At the highest levels of organization such functions as perception, memory, learning and complex behavioral acts call for coordination of many centers. In this coordination a microcircuit and a macrocircuit are not separate entities; one is indissolubly embedded in the other. Thus whether one starts with an interest in the biochemical aspects of behavior or in the electrical activity that underlies it one is led to a common focus: the synaptic circuit through dendrites. Although an understanding of the neural basis of behavior is still remote, some clues have now been given, and it seems clear where the search is most likely to be rewarded.